

ShoeGenAI: A Creativity Support Tool for High-Feasible Shoe Product Design

Hui-Jun Kim
Dong-eui University
Busan, Republic of Korea
kymy8016@gmail.com

Minbong Lee
Kristin Company
Seoul, Republic of Korea
bong@kristinkorea.com

Jeongho Kim
KAIST
Seoul, Republic of Korea
rlawjdghek@kaist.ac.kr

Jaegul Choo
KAIST
Daejeon, Republic of Korea
jchoo@kaist.ac.kr

Sohyun Jeong
KAIST
Seongnam-si, Republic of Korea
jsh0212@kaist.ac.kr

Sung-Hee Kim
Dong-eui University
Busan, Republic of Korea
sh.kim@deu.ac.kr

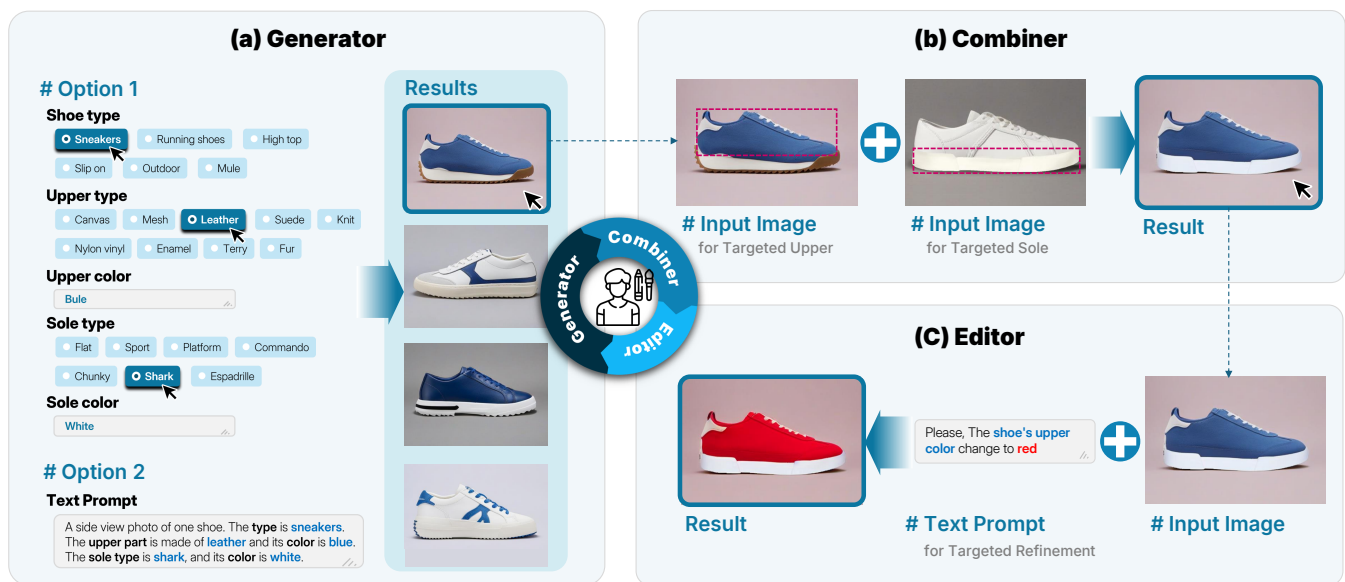


Figure 1: We have developed ShoeGenAI, which consists of three main components: (a) The generator component creates shoe images based on selected keywords from a template or entered text prompts from the users. (b) The combiner component allows the user to blend the upper and sole of two different shoe images. (c) The editor component helps users refine specific aspects of the design through text prompts.

Abstract

Product designers often use generative AI for concept images, but these outputs often lack manufacturing feasibility, requiring repeated adjustments to meet design intentions. Focusing on sneaker design, this study introduces ShoeGenAI, an AI tool that enhances designers' creativity while ensuring feasible outcomes and reducing the need for post-processing. A study with four shoe designers

revealed challenges with both conventional methods and generative AI, leading to four key functions such as model fine-tuning, template-based prompting, combinational creativity support, and targeted refinement. A follow-up study with 20 designers indicated that with ShoeGenAI, they would expect to easily express their intentions to the system, be able to work efficiently with fewer post-processing adjustments, and be satisfied with the practical results. We also discuss differences between professionals and novices using creativity support tools and different types of design tasks, such as replicating a target image and designing from scratch.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3721204>

CCS Concepts

• Human-centered computing → Interactive systems and tools.

Keywords

High-feasibility, Generative Model, Creativity Support, Fashion/Clothing, Prototyping/Implementation

ACM Reference Format:

Hui-Jun Kim, Jeongho Kim, Sohyun Jeong, Minbong Lee, Jaegul Choo, and Sung-Hee Kim. 2025. ShoeGenAI: A Creativity Support Tool for High-Feasible Shoe Product Design. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3706599.3721204>

1 Introduction

Successful product design demands innovative creativity, integrating divergent and convergent thinking [18, 28, 39]. Designers leverage a variety of processes [5, 12, 14, 22, 33, 40], using combinatorial creativity to generate new ideas by mixing reference materials [6, 16]. In such creative work, designers often face mental stress [30, 51] that can lead to design fixation, where they become overly attached to existing ideas. To mitigate this, designers are now turning to generative AI (GenAI) tools like ChatGPT, Dall-E, Midjourney, and Adobe Firefly to expedite the design process and enhance the generation of diverse options.

Within the human-computer interaction (HCI) community, significant research on creativity support tools (CST) extends beyond general services to offer specialized support across various domains. Existing CST research utilizes high-capability large language models, text-to-image, and image-to-image models to enhance usability and creative support across multiple design stages. Most of these tools have supported early stages of design, such as refining prompt queries, recombining elements of reference images, editing specific aspects, and evaluating diagrams [9, 17, 20, 27, 32, 43, 50].

However, research on CSTs that provides feasible outcomes for product designers, particularly within manufacturing, remains limited. High-feasibility designs are practical, and realizable in terms of manufacturing, cost, and other constraints. While existing commercial CST services support creative divergent thinking by providing various concept images [8, 29, 41], the low practical feasibility of generated results often requires designers to engage in labor-intensive post-processing [10]. Additionally, impractical designs with physically impossible structures make manufacturing infeasible, forcing designers to redesign or start from scratch.

To tackle these issues, we concentrated on sneaker design, a field requiring highly feasible designs due to its manufacturing complexity. We propose ShoeGenAI, an AI-driven CST for shoes that ensures high feasibility from a manufacturing perspective, supporting creativity and minimizing post-processing for production. Sneaker design involves iterative stages like ideation, sketching, and sample production, often repeated to achieve the desired final product. While previous studies have emphasized trend analysis, market demand, and sketch refinement in fashion design [20, 44, 47, 48], they have not focused on generating high-feasibility concept images or the creativity support from such results.

Building on CST research highlighting priority differences between novices and professionals [32], we conducted a formative study with four shoe designers to identify key features. This study

revealed four major challenges designers encounter when creating shoe concept images using conventional generative AI tools. These challenges informed the development of four design goals for ShoeGenAI: (1) generating high-feasibility images, (2) providing template-based prompts that integrate sneaker-specific design elements, (3) enabling automated synthesis of design elements, and (4) supporting targeted refinement for modifying specific areas while preserving overall design consistency. We conducted a user study on our tool that involved 10 professional and 10 novice designers conducting a shoe image replication task. The results suggest that ShoeGenAI enhances feasibility and reduces post-processing of shoe concept images. Additionally, we analyzed the effect of ShoeGenAI on the product design process, discussing the differences between professionals and novices working with CSTs and considering the CST's role for product designers.

2 Related Works

2.1 GenAI's potential to support product designer

Recent advances in diffusion techniques [31, 37, 38] and CLIP embeddings [36] have streamlined text-to-image generation. This innovation has led to the rise of text-to-image tools like Midjourney, DALL-E, and Stable Diffusion, enabling designers to move beyond traditional visual searches [21, 46, 49] and use commercial GenAI services to visualize concept ideas [8, 11, 29, 41]. These tools boost creativity and efficiency with rapid image generation.

However, limitations remain in understanding design intent in specialized domains. For example, Jin et al. highlighted the need for systems that better express domain-specific design intent when working with fashion designers [41]. Similarly, user studies have raised concerns about the quality and relevance of commercial GenAI outputs, recommending improvements to address these gaps [3, 13]. Challenges in GenAI's practicality and domain-specific support were also identified in studies on user experience design practices. These findings underscore the need for GenAI tools to evolve into intuitive, visually-oriented, and collaborative systems [25]. Additionally, a shift is needed from AI-centered interactions to designer-centered interactions. This requires practical support through multi-modal input methods that consider designers' AI literacy, intuitive input mechanisms for easy communication of design intentions, and workflow designs that account for the specific characteristics of the design process in the given domain [34, 45].

2.2 AI-driven creativity support tools

The HCI community has been advancing AI-driven CSTs to address the limitations of general-purpose GenAIs in enhancing designers' creativity. These tools primarily focus on early design stages, refining user prompts [17, 43], conducting market research, setting design goals [47], enabling rapid prototyping [27], and facilitating interactive edits and style suggestions [20]. They also support generating and recombining reference images to better convey design intent [32, 43]. In automotive design, Chen et al. proposed AutoSpark, which utilizes a Generative AI-driven Kansei Engineering engine powered by semantic networks to assist designers in aligning emotional requirements, expressing design intent, and

enabling rapid prototyping. However, it still falls short of meeting the nuanced demands of product design within this domain [7]. Similarly, in fashion, Han et al. introduced FashionQ, an AI-driven CST designed to help designers find reference images and combine attributes through machine learning. While it improved accessibility to references, it faced challenges in generating novel visuals or effectively expressing design intent [20]. Furthermore, the commercial Generative AI tool Vizcom has shown promise in automotive, footwear, and furniture design, delivering high-quality image results and partial refinement support. However, it remains specialized for specific domain workflows, offering simplified interactions while lacking consideration for feasibility from a manufacturing perspective [1]. Despite these advancements, research on CSTs capable of creating high-feasibility designs—those that are creative, practical, and manufacturable remains limited, especially in fields like furniture, fashion, and home appliances. Addressing these gaps could unlock new possibilities for balancing creativity and practicality in design.

3 Formative Study

To understand the shoe design process and how general shoe designers utilize GenAI, we conducted a preliminary study with four designers (2 professionals and 2 novices). We recruited formative study participants via a recruitment notice on a design community platform, exclusively selecting shoe design majors. Initially, we interviewed them about the existing shoe design process (see Appendix A.1). Then, we asked them to generate initial shoe images using commercial GenAI tools and modify the shape, color, and material of the shoes. During the generation process, we employed the think-aloud method and conducted interviews afterward to gather feedback on the results, limitations of the existing GenAI, and suggestions for improvement. Through this, we identified four major challenges that our system needs to address: **(Challenge 1)** Low feasibility of generated results, requiring extensive rework for product manufacturing, **(Challenge 2)** Difficulty in expressing the intent of shoe design in text, **(Challenge 3)** Labor-intensive processes required to exercise combinational creativity, **(Challenge 4)** The system's low accuracy in understanding the user's design intent during the image modification process, necessitating repeated regenerations following prompt adjustments. Both professionals and novices acknowledged these challenges, but their priorities differed. Novices particularly emphasized the difficulty of clarifying their design intent and finding relevant images, whereas professionals highlighted the challenge of acquiring a diverse array of reference images.

To address these challenges and not only make it easier to find reference images that align with design intent but also provide practical support for concept image generation, we introduced four design goals (DGs) in ShoeGenAI: **(DG 1)** High-feasibility shoe concept images generated using a fine-tuned text-to-image model, **(DG 2)** Template-Based Prompt providing sneaker styles, soles, colors, materials, and other appearance element, **(DG 3)** Automated element synthesis using a fine-tuned image-to-image model, **(DG 4)** Targeted refinement for modifying specific areas while maintaining consistency using a multi-modal model [4].

4 ShoeGenAI

ShoeGenAI is an automatic tool to support the creation of a new concept shoe image. We developed it by incorporating the design process identified in our formative study, which includes reference image exploration, preliminary sketching, 2D visualization, and component modifications to improve manufacturability, followed by iterative refinements incorporating these steps. It generates a range of high-feasibility images to help the designers reflect their intended design goals to reduce the iterative labor-intensive process of combining or modifying concept images. ShoeGenAI is a web-based system built using gradio. We used a machine with an Intel(R) Xeon(R) Silver 4214 Core Processor and NVIDIA GeForce TITAN RTX to implement ShoeGenAI.

4.1 Technical Detail

4.1.1 Generator. The first component of ShoeGenAI is Generator, a GenAI that generates shoe images based on the given text prompts, tackling challenge 2: the difficulty in expressing the intent of shoe design in text. While recent generative AI models can generate general shoes images, they lack the capacity to understand specialized terminology related to shoes. Additionally, the shoe images generated by these models frequently exhibit inconsistencies, such as varying perspectives or the inclusion of extraneous objects like human body parts, which are unnecessary and inefficient for product design (see Appendix A.2 Figure 4).

To address this limitation, we build well-designed text prompts with precise descriptors. The prompts follow a standardized format: "A side view photo of one shoe. The type is [shoe type]. The upper part is made of [upper part material]. The sole type is [sole type]." These prompts generated based on human-labeled data, providing the model with clear and specific design instructions. Moreover, the prompts enable the model to generate images that accurately capture essential design features. Using these text prompts, we fine-tune SDXL [35] with LoRA [19] and subsequently evaluate the model. During user inference, we include color options for the upper part and the sole to the text prompts, which differs from the training phase, as we believe the diffusion model's prior knowledge of color can be leveraged.

4.1.2 Combiner. The second component is the Combiner, a generative AI that blends the upper and sole of two different shoe images, tackling challenge 3: labor-intensive processes involved in exercising combinational creativity. As shown in Figure 1 (b), our combiner produced high-feasibility synthetic images when given arbitrary shoe and sole images.

For training the Combiner, we first separate the given shoe images and reconstruct them based on Stable Diffusion [38]. In separate stage, we utilize a pre-trained segmentation model, SAM [24], which segments the clicked image into individual objects, to separate the upper and sole part of the shoe. The separated sole is resized to a uniform size, agnostic to its size information. We apply a brightness perturbation to the resized sole image to train the AI model to implicitly learn the relative brightness differences intrinsically, resulting in high-fidelity outputs. Then, we finetune the model to reconstruct the original shoe image using the masked upper image and the perturbed sole image as a reference [23]. Therefore, users could easily combine their chosen shoe image and sole image

through the combiner. Examples of the outputs generated by our Combiner can be found in the Appendix A.2 Figure 5.

4.1.3 Editor. The third component of ShoeGenAI is the Editor, a generative AI that edits concept images from designer text prompts for target refinement, tackling low accuracy in understanding the design intent (Challenge 4). Our editor produces high-consistency target refinement images except for target regions. To implement the editor, we decided to use the InstructPix2Pix [4] model because of its ability to maintain consistency with the original shoe features except for the target region during editing. This allowed users to edit the concept image to any desired color or material while maintaining a relatively high level of consistency.

5 User Study

In this section, we evaluate the effectiveness of ShoeGenAI through a comprehensive user study. To this end, we enabled designers to collaborate with GenAI in the iterative processes traditionally required for prototype development, including gathering client requirements, conducting reference searches, sketching, and realization. The study involved a task where participants were asked to conceptually replicate a shoe image. Based on findings from our formative study, we presented eight candidate photos from various sneaker categories (see Appendix A.3 Figure 6) to minimize the influence of participants' individual experiences on tool usage. The scenario given for the task was: "A client provides an image and requests you to design a similar shoe using it." The task was designed following the concept image design process identified in our formative study. It also incorporated the approach used by Jeon et al. [20], which presents a design goal image and contextualizes the participant's situation to reduce the impact of personal design experience. The baseline chosen for comparison was Dall-E 3 [15], which received the highest evaluation in concept image generation in the existing product designs among commercial GenAIs used in formative research [11]. Additionally, the final results generated by the participants can be found in the Appendix A.3 Figure 7.

5.1 Participants

In the formal study, we identified that the challenges faced by shoe designers differ based on their experience level. To explore this further, we conducted a user study involving 20 shoe designers (12 women and 8 men) who were recruited from a designer community and whose primary major is shoe design. The participants included 10 professionals (3 women, 7 men; P1-P10) with over two years of experience in the shoe industry and 10 novices (9 women, 1 man; P11-P20) currently enrolled in undergraduate programs specializing in shoe design.

5.2 Study Procedure

The study lasted 1 hour, beginning with an introduction and informed consent. It was conducted both in-person and online via Zoom for remote participants. After a 5-minute explanation of the tools (Baseline and ShoeGenAI) and protocol, participants were instructed to create concept images similar to those provided for manufacturing. They selected an image and were briefed on the scenario before spending 20 minutes designing a shoe concept. To avoid ordering effects, the conditions were counterbalanced,

with a 5-minute break between them. After the task, a 20-minute semi-structured interview explored their experiences. Participants received 100,000 KRW (75 USD) as compensation.

5.3 Measures

Questionnaire: For measures, the surveys conducted after each task included questions that asked participants to rate on a 7-point Likert scale their satisfaction with the result and the design process [32, 43], the perceived difficulty during design using these tools, efficiency with these, creativity supported from these, and feasibility of the concept image from these. The questionnaire items for each metric are as shown in Appendix A.3 Table 1.

Interview: In the in-depth interviews, participants were asked about perceived differences between groups regarding the feasibility of their designed shoe concept images. Questions also explored differences in expressing design intentions, the impact on their design processes, and the potential use of tools for creative work versus repetitive tasks. Additionally, participants were asked about differences from the perceived baseline during the design process, GenAI's capabilities and perceptions as shoe designers, and the features needed to enhance creativity.

6 Results

To determine statistical significance, a Wilcoxon signed-rank test was conducted with a significance level set at $p < 0.05$. The survey ratings were analyzed to observe the effect of ShoeGenAI on the perceived experience of the designers, namely satisfaction, perceived difficulty, efficiency, and creativity. This non-parametric test was chosen because the data were not normally distributed. Although it is a non-parametric test, we also reported the mean and standard deviation to help understand the trends in the ratings. We observed and compared the perceived user experience of each tool presented according to the differences in the participants' design experience levels. And through the results of in-depth interviews, we found the cause of the perceived experience. Our results indicate a clear trend that ShoeGenAI provides better support for the shoe design process compared to the baseline, with this effect being more pronounced among professionals.

Satisfaction: Questions 1 through 4 (Q1–Q4) assessed participants' satisfaction with the output image from each tool. As shown in Figure 2(a), professionals rated ShoeGenAI higher than the baseline for all questions, while novices only rated Q4 higher than the baseline. The results for each question are detailed below.

For overall satisfaction (Q1), novices showed no statistically significant difference between the two tools ($Z = -0.53, p = .60$), with the baseline ($M = 4.70, SD = 1.83$) and ShoeGenAI ($M = 5.20, SD = 1.75$). However, professionals showed a significant difference ($Z = -2.09, p = .04$), with ShoeGenAI ($M = 5.10, SD = 1.66$) rated higher than the baseline ($M = 3.50, SD = 1.90$). For overall quality (Q2), novices showed no statistically significant difference ($Z = -0.69, p = .49$), with the baseline ($M = 4.90, SD = 1.60$) and ShoeGenAI ($M = 5.50, SD = 1.08$). However, professionals showed a significant difference ($Z = -2.54, p = .01$), with ShoeGenAI ($M = 4.80, SD = 1.81$) rated higher than the baseline ($M = 3.00, SD = 1.49$).

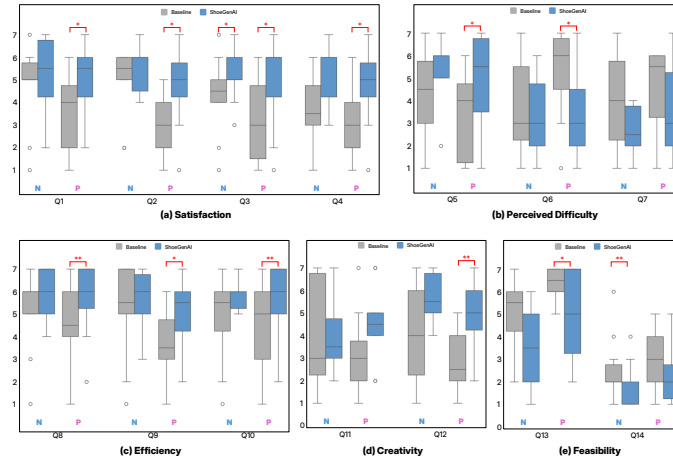


Figure 2: Questionnaire results from T1. The difference between novices (N) and professionals (P) in their experience of designing shoe concept images using the given tools, based on replicating a targeted image. The number following "Q" in the figure refers to the corresponding question number, and the questions can be found in Appendix. The significance levels are as follows: * $p < 0.05$, ** $p < 0.01$.

For freedom to express design intentions (Q3), novices showed no significant difference ($Z = -1.19$, $p = .23$), with the baseline ($M = 4.30$, $SD = 1.77$) and ShoeGenAI ($M = 3.20$, $SD = 1.14$). However, professionals showed a significant difference ($Z = -2.46$, $p = .01$), with ShoeGenAI ($M = 4.90$, $SD = 1.97$) rated higher than the baseline ($M = 3.20$, $SD = 1.81$). Lastly, for accurate expression of design intent (Q4), novices showed a significant difference ($Z = -2.21$, $p = .03$), with ShoeGenAI ($M = 5.40$, $SD = 1.35$) rated higher than the baseline ($M = 3.70$, $SD = 1.64$). Similarly, professionals also showed a significant difference ($Z = -2.46$, $p = .01$), with ShoeGenAI ($M = 4.70$, $SD = 1.70$) rated higher than the baseline ($M = 3.10$, $SD = 1.66$).

Perceived Difficulty: Questions 5 through 7 (Q5–Q7) measured perceived difficulty using the tools, as illustrated in Figure 2(b). A higher score on Q6 and Q7 indicates greater difficulty, while a lower score on Q5 indicates greater difficulty. For Q5, which asked if the tool understood user intentions well, novices showed no statistically significant difference ($Z = -1.45$, $p = .15$), with the baseline ($M = 4.20$, $SD = 2.20$) and ShoeGenAI ($M = 5.50$, $SD = 1.43$). However, professionals showed a significant difference ($Z = -2.46$, $p = .01$), with ShoeGenAI ($M = 5.00$, $SD = 2.05$) rated higher than the baseline ($M = 3.30$, $SD = 1.89$).

For Q6, which assessed task difficulty, novices showed no significant difference ($Z = -0.72$, $p = .47$), with the baseline ($M = 3.80$, $SD = 2.15$) and ShoeGenAI ($M = 3.20$, $SD = 1.75$). Professionals also showed no significant difference ($Z = -2.39$, $p = .02$), but rated ShoeGenAI ($M = 3.40$, $SD = 1.96$) as less difficult than the baseline ($M = 5.30$, $SD = 2.00$). Lastly, for Q7 (ease of use), novices showed no significant difference ($Z = -1.71$, $p = .09$), with the baseline ($M = 4.00$, $SD = 2.05$) and ShoeGenAI ($M = 2.60$, $SD = 1.17$). Professionals also showed no significant difference ($Z = -1.13$, $p = .26$), with the baseline ($M = 4.60$, $SD = 1.78$) and ShoeGenAI ($M = 3.60$, $SD = 2.22$).

Efficiency: Questions 8 through 10 (Q8–Q10) assessed efficiency, as shown in Figure 2(c). A higher score indicates greater efficiency. Professionals rated ShoeGenAI higher across all questions, while novices showed no significant differences. For Q8 (shortening work time), novices showed no statistically significant difference ($Z = -1.28$, $p = .20$), with the baseline ($M = 5.20$, $SD = 1.87$) and ShoeGenAI ($M = 5.90$, $SD = 1.20$). Professionals, however, showed a significant difference ($Z = -2.81$, $p = .01$), with ShoeGenAI ($M = 5.70$, $SD = 1.64$) rated higher than the baseline ($M = 4.60$, $SD = 1.78$).

For Q9 (assisting client communication), novices showed no significant difference ($Z = -0.51$, $p = .61$), with the baseline ($M = 5.20$, $SD = 2.15$) and ShoeGenAI ($M = 5.70$, $SD = 1.25$). Professionals showed a significant difference ($Z = -2.57$, $p = .01$), with ShoeGenAI ($M = 5.10$, $SD = 1.66$) rated higher than the baseline ($M = 3.60$, $SD = 1.51$). Lastly, for Q10 (reducing repetitive labor), novices showed no statistically significant difference ($Z = -1.40$, $p = .16$), with the baseline ($M = 4.90$, $SD = 2.02$) and ShoeGenAI ($M = 5.80$, $SD = 0.63$). Professionals, however, showed a significant difference ($Z = -2.64$, $p = .01$), with ShoeGenAI ($M = 5.60$, $SD = 1.65$) rated higher than the baseline ($M = 4.30$, $SD = 2.16$).

Creativity: Questions 11 and 12 (Q11–Q12) measured creativity, as shown in Figure 2(d). A higher score indicates greater creativity support.

For Q11 (originality of output), novices showed no significant difference ($Z = -0.86$, $p = .93$), with the baseline ($M = 4.10$, $SD = 2.38$) and ShoeGenAI ($M = 4.00$, $SD = 1.56$). Similarly, professionals showed no significant difference ($Z = -1.70$, $p = .09$), with the baseline ($M = 3.20$, $SD = 1.75$) and ShoeGenAI ($M = 4.50$, $SD = 1.72$). For Q12 (concretizing design intent), novices showed no significant difference ($Z = -1.41$, $p = .16$), with the baseline ($M = 4.10$, $SD = 2.08$) and ShoeGenAI ($M = 5.60$, $SD = 1.17$). Professionals, however, showed a significant difference ($Z = -2.87$,

$p = .004$), with ShoeGenAI ($M = 4.80$, $SD = 1.69$) rated higher than the baseline ($M = 2.90$, $SD = 1.52$).

Feasibility: As shown in Figure 2(e), Questions 13 and 14 assessed the feasibility of the output images for production. A lower score indicates that the output image is more feasible for production.

For Q13, which asked whether users felt post-processing was needed for the output image, novices showed no statistically significant difference between the tools ($Z = -1.80$, $p = .07$), with the baseline ($M = 5.00$, $SD = 1.83$) and ShoeGenAI ($M = 3.50$, $SD = 1.72$). However, professionals showed a statistically significant difference ($Z = -2.12$, $p = .03$), with ShoeGenAI ($M = 5.00$, $SD = 1.94$) rated lower than the baseline ($M = 6.40$, $SD = 0.70$). For Q14, which asked whether the image contained non-existing materials (where a lower score indicates higher feasibility), novices showed a statistically significant difference between the tools ($Z = -2.71$, $p = .01$), with ShoeGenAI ($M = 1.70$, $SD = 1.06$) rated lower than the baseline ($M = 2.60$, $SD = 1.43$). However, professionals showed no statistically significant difference between the tools ($Z = -1.17$, $p = .24$), with the baseline ($M = 3.00$, $SD = 1.56$) and ShoeGenAI ($M = 2.30$, $SD = 1.34$).

7 Discussion

7.1 Professionals vs. Novices

Through the results of the user study, we identified notable differences in how professionals and novices assess the feasibility of a product based on their design experience. Specifically, professionals were found to be more sensitive than novices in evaluating the feasibility of the outcome, as supported by Q13. During interviews, these differences became more evident. Novices primarily focused on form related elements of the shoe, reporting issues that would require correction in the post-processing stage, such as the “presence or absence of shoelace holes” or “the shoelaces being integrated into the shoe’s body.” In contrast, professionals not only addressed form-related elements but also raised concerns about material composition and structural design. For example, one professional remarked, “*It is structurally impossible to place leather material over mesh in the upper part of the shoe*” (P11), while another noted, “*The pattern on the mesh is irregular*” (P20). Professionals also highlighted the proportions between shoe components, such as “*It is difficult to manufacture a sole that protrudes more than the upper part*” (P13), demonstrating a more detailed analysis of aspects requiring post-processing modifications.

There were also differences in how the two groups evaluated the tool’s efficiency, as supported by Q8. Professionals valued the ability to quickly generate a variety of high-feasibility concept images within the same category, noting that it could significantly reduce overall design time. For instance, one professional commented, “*It’s nice that new designs keep coming out just by clicking, without needing additional work*.” (P14). Additionally, professionals highly appreciated the tool’s functionality for automating repetitive tasks, such as combining design elements. Examples include remarks like “*It’s convenient that the background is automatically removed to combine the sole and upper*” (P11) and “*It’s fast, so it would be great if this feature were commercialized*” (P20). Some professionals (P11 and P20) also discussed challenges related to repetitive tasks

in design work, aligning with the results of the formative study. While novices generally provided positive evaluations regarding the efficiency of ShoeGenAI, they did not express that the tool could reduce the overall design process time. This indicates a divergence in the perceived impact of the tool’s efficiency between the two groups.

Our findings suggest that ShoeGenAI effectively supports novices who have relatively less expertise—by generating concept images that reflect practical requirements in the field, thanks to its integration of design workflows and high feasibility from a manufacturing perspective. Additionally, by leveraging a simple generation method based on key domain-specific keywords, ShoeGenAI helps clarify design intent more effectively. Moreover, the detailed design elements in the generated images met the expectations of professionals, demonstrating its potential to significantly reduce design time. This aligns with previous research on essential features of CSTs, which emphasize the importance of deep domain understanding [2]. ShoeGenAI not only captures and satisfies the implicit and detailed knowledge of professionals but also compensates for knowledge gaps, fulfilling a key requirement highlighted in prior studies [42].

7.2 Technical limitations and implications for better AI-driven CST design

ShoeGenAI includes three core modules: the Generator, Combiner, and Editor. The Combiner and Editor allow users to customize specific shoe parts, such as the sole or upper, using segmentation techniques. We utilized pre-trained segmentation models [24] and attention maps [4], fine-tuned for the shoe domain to enhance understanding of shoe-specific terminology and maintain strong generalization from large-scale datasets. Future improvements could focus on creating shoe-specific segmentation labels and using guided loss [26] to improve text alignment and design accuracy.

As noted in the formative study, professionals often overcome design fixation by recombining elements from multiple reference images. While ShoeGenAI currently combines shoe elements via text prompts, further development is needed to handle more complex combinations and integrate elements from multiple concept images. Expanding the system to extract and reassemble sub-components from shoe images would better support detailed and high-feasibility designs.

8 Conclusion and Future Work

We present the AI-driven CST, ShoeGenAI, designed to deliver high-feasibility concept images specifically for shoe designers. A formative study revealed that designers face challenges with repetitive labor and ideation during shoe production. Commercial GenAI tools often produce low-feasibility results, requiring additional post-processing and increasing workload instead of alleviating it.

In a user study with professionals and novices, ShoeGenAI addressed these issues by generating high-feasibility designs and improving efficiency. The study highlighted differences based on design experience: professionals showed greater sensitivity to feasibility and higher satisfaction with outcomes, while also benefiting more from reduced repetitive tasks. These findings position high-feasibility as a key criterion for future CSTs targeting product designers.

User study feedback suggested additional features to expand creative support and improve outcomes. For combinational creativity, generating diverse aesthetic images that preserve shoe structure is essential. Users also expressed a desire for freeform text inputs to better convey their intentions, which will require enhanced models and datasets. Finally, ensuring feasibility from a materials perspective will demand advancements in generating high-resolution, real-world textures.

Acknowledgments

This work was partly supported by the Institute of Information & Communications Technology Planning & Evaluation(IITP)-Innovative Human Resource Development for Local Intellectualization program grant funded by the Korea government(MSIT)(IITP-2025-RS-2020-II201791, 50%) and Technological Innovation R&D Program (RS-2024-00438252, 50%) funded by the Ministry of SMEs and Startups(MSS, Korea) award number(s).

References

- [1] Regina Aguirre. 2023. Regina Aguirre: Pioneering the Intersection of AI and Design with Vizcom. <https://www.vizcom.ai/blog/regina-aguirre-the-first-higher-education-student-to-become-a-vizcom-ambassador> Accessed on 2 March, 2025.
- [2] Matthew T Beaudouin-Lafon, Jane L E, and Haijun Xia. 2023. Color Field: Developing Professional Vision by Visualizing the Effects of Color Filters. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 101, 16 pages. doi:10.1145/3586183.3606828
- [3] Ross Brisco, Laura Hay, and Sam Dhami. 2023. Exploring the role of text-to-image AI in concept generation. *Proceedings of the Design Society* 3 (2023), 1835–1844.
- [4] Tim Brooks, Aleksander Holynski, and Alexei A. Efros. 2023. InstructPix2Pix: Learning to Follow Image Editing Instructions. In *CVPR*.
- [5] Tony Buzan, Barry Buzan, and James Harrison. 2010. The mind map book: Unlock your creativity, boost your memory, change your life. (No Title) (2010).
- [6] Donald T Campbell. 1960. Blind variation and selective retentions in creative thought as in other knowledge processes. *Psychological review* 67, 6 (1960), 380.
- [7] Liuqing Chen, Qianzhi Jing, Yixin Tsang, Qianyi Wang, Ruocong Liu, Duowei Xia, Yunzhan Zhou, and Lingyun Sun. 2024. AutoSpark: Supporting Automobile Appearance Design Ideation with Kansei Engineering and Generative AI. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) (UIST '24). Association for Computing Machinery, New York, NY, USA, Article 108, 19 pages. doi:10.1145/3654777.3676337
- [8] Shih-Hung Cheng. 2023. Impact of Generative Artificial Intelligence on Footwear Design Concept and Ideation. *Engineering Proceedings* 55, 1 (2023). doi:10.3390/engproc2023055032
- [9] DaEun Choi, Sumin Hong, Jeongeun Park, John Joon Young Chung, and Juho Kim. 2024. CreativeConnect: Supporting Reference Recombination for Graphic Design Ideation with Generative AI. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 1055, 25 pages. doi:10.1145/3613904.3642794
- [10] John Joon Young Chung and Eytan Adar. 2023. PromptPaint: Steering Text-to-Image Generation Through Paint Medium-like Interactions. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 6, 17 pages. doi:10.1145/3586183.3606777
- [11] Sam Dhami, Ross Brisco, et al. 2024. A COMPARISON OF ARTIFICIAL INTELLIGENCE IMAGE GENERATION TOOLS IN PRODUCT DESIGN. In *DS 131: Proceedings of the International Conference on Engineering and Product Design Education (E&PDE 2024)*. 13–18.
- [12] Peter Facione. 1989. Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction. *Research Findings and Recommendations* 315 (11 1989).
- [13] Fabio Antonio Figoli, Lucia Rampino, Francesca Mattioli, et al. 2022. AI in the design process: training the human-AI collaboration. In *Proceedings of the 24th International Conference on Engineering and Product Design Education (E&PDE 2022)*. 1–6.
- [14] Benjamin Franklin. 2022. *Autobiography of Benjamin Franklin*. Project Gutenberg.
- [15] Li Jing Aditya Ramesh Gabriel Goh, James Betker. 2024. Dall-e 3. <https://openai.com/index/dall-e-3/> Accessed on 24 February, 2025.
- [16] Ji Han, Dongmyung Park, Feng Shi, Liuqing Chen, Min Hua, and Peter RN Childs. 2019. Three driven approaches to combinational creativity: Problem-, similarity- and inspiration-driven. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science* 233, 2 (2019), 373–384.
- [17] Yihan Hou, Manling Yang, Hao Cui, Lei Wang, Jie Xu, and Wei Zeng. 2024. C2Ideas: Supporting Creative Interior Color Design Ideation with a Large Language Model. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 172, 18 pages. doi:10.1145/3613904.3642224
- [18] Shih-Wen Hsiao and Jyh-Rong Chou. 2004. A creativity-based design process for innovative product design. *International Journal of Industrial Ergonomics* 34, 5 (2004), 421–443. doi:10.1016/j.ergon.2004.05.005
- [19] Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=nZeVKeeFYf9>
- [20] Youngseung Jeon, Seungwan Jin, Patrick C. Shih, and Kyungsik Han. 2021. FashionQ: An AI-Driven Creativity Support Tool for Facilitating Ideation in Fashion Design. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 576, 18 pages. doi:10.1145/3411764.3445093
- [21] Youwen Kang, Zhida Sun, Sitong Wang, Zeyu Huang, Ziming Wu, and Xiaojuan Ma. 2021. MetaMap: Supporting Visual Metaphor Ideation through Multi-dimensional Example-based Exploration. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 427, 15 pages. doi:10.1145/3411764.3445325
- [22] Tom Kelley. 2001. *The art of innovation: Lessons in creativity from IDEO, America's leading design firm*. Vol. 10. Currency.
- [23] Jeongho Kim, Guojung Gu, Minhoo Park, Sunghyun Park, and Jaegul Choo. 2024. Stableviton: Learning semantic correspondence with latent diffusion model for virtual try-on. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8176–8185.
- [24] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4015–4026.
- [25] Jie Li, Hancheng Cao, Laura Lin, Youyang Hou, Ruihao Zhu, and Abdallah El Ali. 2024. User Experience Design Professionals' Perceptions of Generative Artificial Intelligence. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 381, 18 pages. doi:10.1145/3613904.3642114
- [26] Shikai Li, Jianglin Fu, Kaiyuan Liu, Wentao Wang, Kwan-Yee Lin, and Wayne Wu. 2024. CosmicMan: A Text-to-Image Foundation Model for Humans. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6955–6965.
- [27] David Chuan-En Lin and Nikolas Martelaro. 2024. Jigsaw: Supporting Designers to Prototype Multimodal Applications by Chaining AI Foundation Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 4, 15 pages. doi:10.1145/3613904.3641920
- [28] Soran Malaie, Michael J Spivey, and Tyler Marghetis. 2024. Divergent and convergent creativity are different kinds of foraging. *Psychological Science* (2024), 09567976241245695.
- [29] Nusrat Jahan Mim, Dipannita Nandi, Sadaf Sumyia Khan, Arundhuti Dey, and Syed Ishtiaque Ahmed. 2024. In-Between Visuals and Visible: The Impacts of Text-to-Image Generative AI Tools on Digital Image-making Practices in the Global South. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 474, 18 pages. doi:10.1145/3613904.3641951
- [30] Thanh An Nguyen and Yong Zeng. 2014. A physiological study of relationship between designer's mental effort and mental stress during conceptual design. *Computer-Aided Design* 54 (2014), 3–18. doi:10.1016/j.cad.2013.10.002 APPLICATION OF BRAIN-COMPUTER INTERFACES IN CAD/E SYSTEMS.
- [31] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. 2021. Glide: Towards photorealistic image generation and editing with text-guided diffusion models. *arXiv preprint arXiv:2112.10741* (2021).
- [32] Jeongseok Oh, Seungju Kim, and Seungjun Kim. 2024. LumiMood: A Creativity Support Tool for Designing the Mood of a 3D Scene. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 174, 21 pages. doi:10.1145/3613904.3642440
- [33] AF Osborn. 1963. Principles and procedures of creative problem-solving. *Tokyo: Success Maturation Institute of Japan* (1963).
- [34] Hyerim Park, Joscha Eirich, Andre Luckow, and Michael Sedlmair. 2024. "We Are Visual Thinkers, Not Verbal Thinkers!": A Thematic Analysis of How Professional Designers Use Generative AI Image Generation Tools. In *Proceedings of the 13th*

Nordic Conference on Human-Computer Interaction (Uppsala, Sweden) (*NordCHI '24*). Association for Computing Machinery, New York, NY, USA, Article 35, 14 pages. doi:10.1145/3679318.3685370

- [35] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2024. SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. In *The Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=di52zR8xgf>
- [36] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PMLR, 8748–8763.
- [37] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125* 1, 2 (2022), 3.
- [38] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
- [39] Mark Runco. 2014. Creativity: Theories and Themes: Research, Development, and Practice. *Creativity: Theories and Themes: Research, Development, and Practice* (04 2014). doi:10.1016/C2012-0-06920-7
- [40] Thomas L Saaty. 1980. The analytic hierarchy process (AHP). *The Journal of the Operational Research Society* 41, 11 (1980), 1073–1076.
- [41] YJYJ Self and Kyungho Lee. 2024. Understanding Fashion Designers' Behavior Using Generative AI for Early-Stage Concept Ideation and Revision. *Archives of Design Research-Vol 37*, 3 (2024), 25–45.
- [42] Kihoon Son, DaEun Choi, Tae Soo Kim, and Juho Kim. 2024. Demystifying Tacit Knowledge in Graphic Design: Characteristics, Instances, Approaches, and Guidelines. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 221, 18 pages. doi:10.1145/3613904.3642886
- [43] Kihoon Son, DaEun Choi, Tae Soo Kim, Young-Ho Kim, and Juho Kim. 2024. GenQuery: Supporting Expressive Visual Search with Generative Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 180, 19 pages. doi:10.1145/3613904.3642847
- [44] Jochen Suessmuth, Florian Fick, and Stan Van Der Vossen. 2023. Generative AI for Concept Creation in Footwear Design. In *ACM SIGGRAPH 2023 Talks* (Los Angeles, CA, USA) (*SIGGRAPH '23*). Association for Computing Machinery, New York, NY, USA, Article 17, 2 pages. doi:10.1145/3587421.3595416
- [45] Macy Takaffoli, Sijia Li, and Ville Mäkelä. 2024. Generative AI in User Experience Design and Research: How Do UX Practitioners, Teams, and Companies Use GenAI in Industry?. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (*DIS '24*). Association for Computing Machinery, New York, NY, USA, 1579–1593. doi:10.1145/3643834.3660720
- [46] Hideyuki Tamura and Naokazu Yokoya. 1984. Image database systems: A survey. *Pattern Recognition* 17, 1 (1984), 29–43. doi:10.1016/0031-3203(84)90033-5
- [47] Sambheet Tiady, Anirban Majumder, and Deepak Gupta. 2023. PRODIGY: Product Design Guidance at Scale. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management* (Birmingham, United Kingdom) (*CIKM '23*). Association for Computing Machinery, New York, NY, USA, 4836–4842. doi:10.1145/3583780.3615494
- [48] Di Wu, Zhiwang Yu, Nan Ma, Jianan Jiang, Yuetian Wang, Guixiang Zhou, Hanhui Deng, and Yi Li. 2023. StyleMe: Towards Intelligent Fashion Generation with Designer Style. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (*CHI '23*). Association for Computing Machinery, New York, NY, USA, Article 23, 16 pages. doi:10.1145/3544548.3581377
- [49] Tom Yeh, Kristen Grauman, Konrad Tollmar, and Trevor Darrell. 2005. A picture is worth a thousand keywords: image-based object search on a mobile platform. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems* (Portland, OR, USA) (*CHI EA '05*). Association for Computing Machinery, New York, NY, USA, 2025–2028. doi:10.1145/1056808.1057083
- [50] Ruican Zhong, Donghoon Shin, Rosemary Meza, Predrag Klasnja, Lucas Colusso, and Gary Hsieh. 2024. AI-Assisted Causal Pathway Diagram for Human-Centered Design. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 2, 19 pages. doi:10.1145/3613904.3642179
- [51] Shan Zhu, Shengji Yao, and Yong Zeng. 2007. A novel approach to quantifying designer's mental stress in the conceptual design process. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Vol. 48035. 593–600.

A Appendix

A.1 The traditional design process and the design process with Gen AI

In studying the shoe design processes, we found that the designers often used reference images to clarify their design intentions. They created preliminary sketches by referencing various images found through visual search tools that resembled their target design, leading up to the 2D realization stage. They produced multiple concept images during this phase by experimenting with colors, materials, and combinations of shoe components like the upper and sole. The feasibility of these designs was then evaluated from a manufacturing standpoint, requiring iterative modifications and verification to improve their production viability (see Figure 3).

A.2 ShoeGenAI's performance

This section provides additional information about the results generated through ShoeGenAI. ShoeGenAI offers three modules, but it only provides example outputs for the Generator (see Figure 4) and Combiner (see Figure 5) modules, which have undergone fine-tuning.

A.3 Components of User Study and Result Images

This section provides detailed information about the materials used in the User Study conducted in this research. It includes Figure 6 of the eight sneaker candidates presented to participants, followed by a Table 1 showing evaluation metrics and questions of questionnaire. Additionally, we provide some of the results generated directly by participants in our user study (see Figure 7).

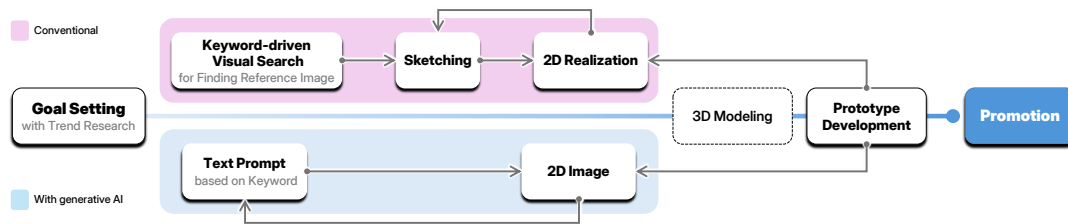


Figure 3: Shoe Design Process Flow: This flow represents the shoe design process identified through formative research. The top pink box represents the conventional design process, while the bottom blue box illustrates the process when generative AI was used. The forward arrows indicate the concept image design phase, while the reverse arrows represent modifications made to incorporate feedback. The dotted line signifies an optional process mentioned by only a few designers.

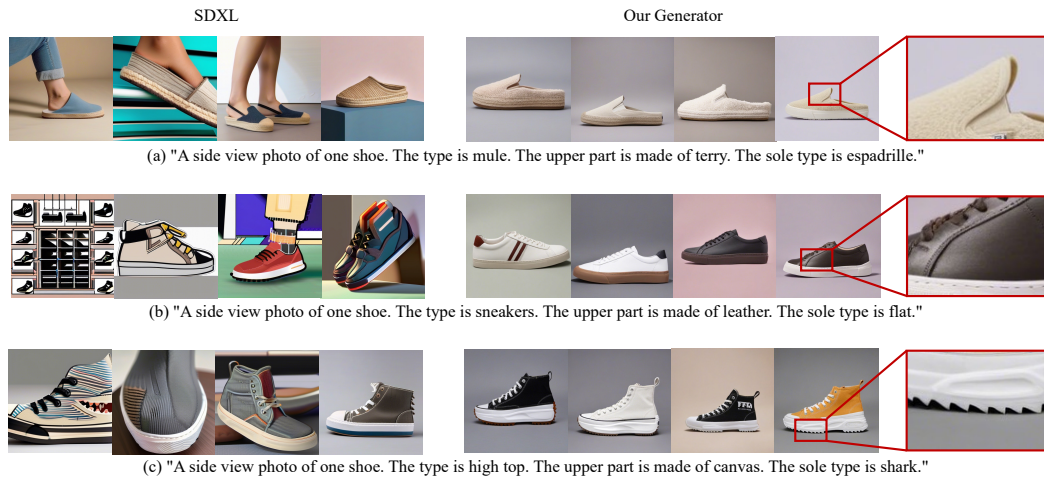


Figure 4: Qualitative comparison of generated images between the baseline model, SDXL (left four images), and our Generator (right four images). The images are generated using text prompts below them. This result shows Generator's ability to more accurately interpret the prompts and produce visually coherent results.

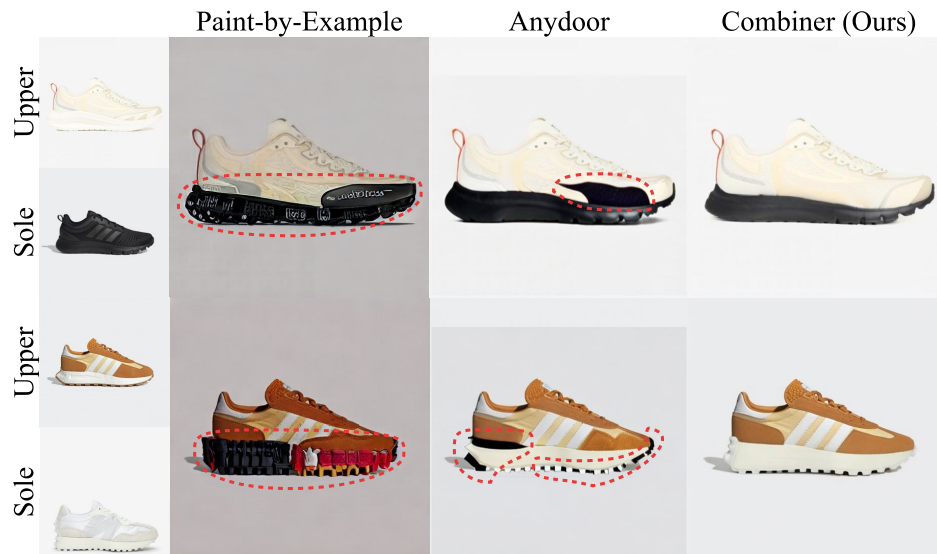


Figure 5: The actual examples of new shoe designs where the upper and sole are combined. Our Combiner on the far right demonstrates high synthesis accuracy and seamless results.

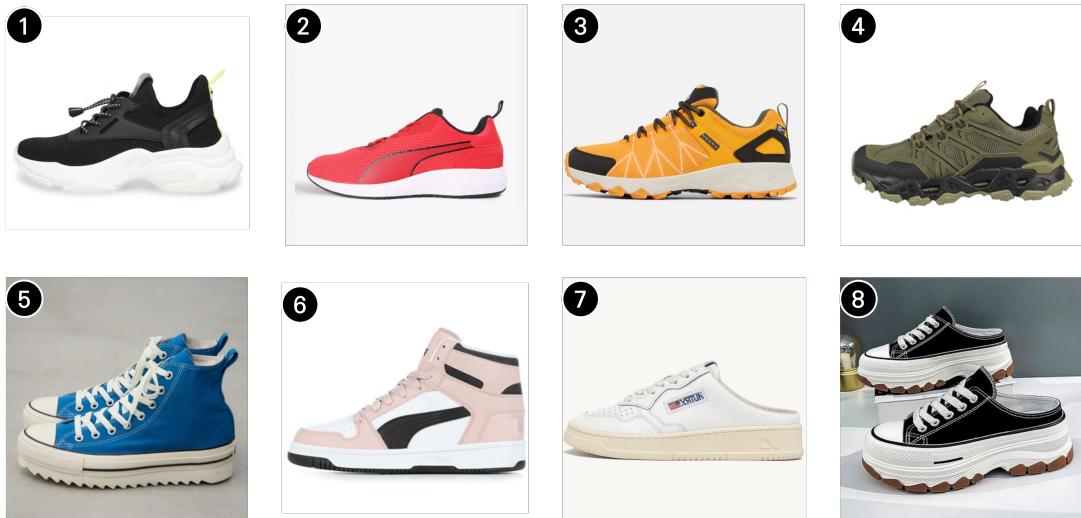


Figure 6: User study task shoe photo candidates. These are 8 sneaker-category shoe photos presented to the participants. The participants selected one of these and performed. The numbers in the upper left corner of each picture indicate the identification code.

Table 1: User Study Questionnaire. Participants were asked about satisfaction with the design outcome, perceived difficulty, efficiency, creativity, and feasibility.

Metrics	Item	Question
Satisfaction	1	I am generally satisfied with the results found through this system.
	2	I believe the quality of the results obtained through this system is high enough to be manufactured in a factory.
	3	I was able to freely express my design intentions in this systems.
	4	I was able to express my design intentions in this system accurately.
Perceived Difficulty	5	The system understood my design intentions well.
	6	It was challenging to modify the concept image to resemble the target shoe.
	7	It was challenging to express my design intent in input fields into this system.
Efficiency	8	I think that with this system, the shoe design time will be shortened.
	9	I believe that through the results of this system, communication with clients will become easier.
	10	I believe that through this system, we will be able to reduce the simple repetitive labor that occurs during shoe design.
Creativity	11	The concept image through this system was original.
	12	Through this system, I was able to concretize my design intent.
Feasibility	13	I think a lot of post-processing work will be needed to produce the results of this system into shoes.
	14	The results of this system are believed to contain materials that do not exist in the real world.

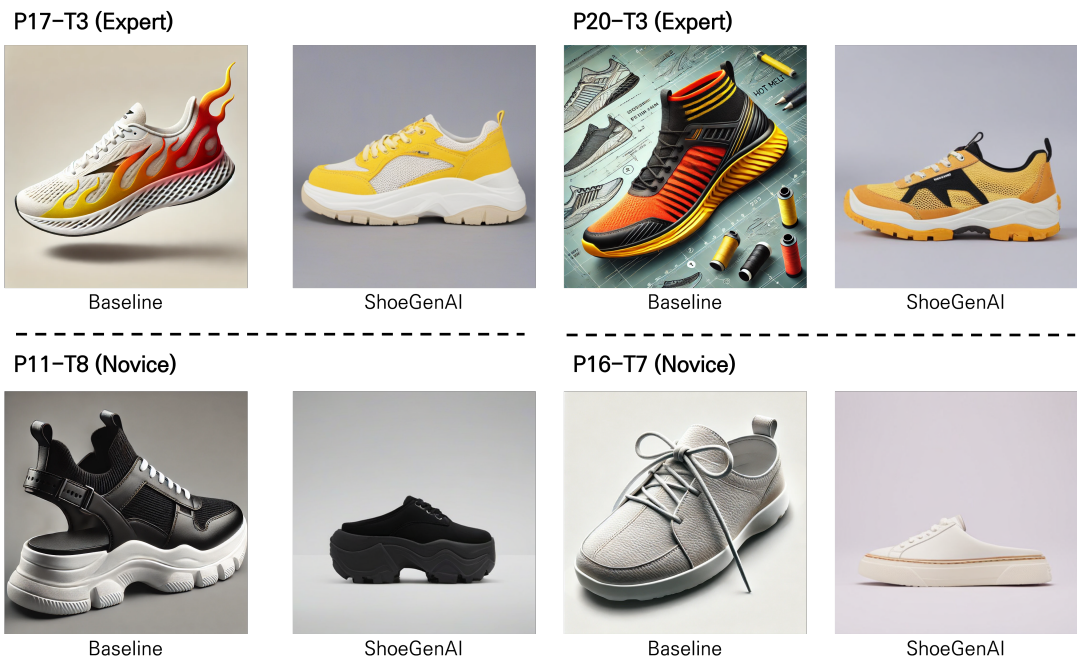


Figure 7: The final results of generating images using ShoeGenAI and baseline for each participant in the user study. The note P17-T3 refers to the image generated by participant number 17 when target image 3 was set as the target.